

Information Visualization

2D Visualization: Spatial Data & Relationships

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Learning Objectives

By the end of this lecture, you will be able to:

- ✓ Identify correlations in scatterplots and interpret the Pearson correlation coefficient
- ✓ Build heatmaps with perceptually uniform palettes and avoid misleading rainbow scales
- ✓ Design spatial layouts using treemaps, chord diagrams, force-directed graphs, and small multiples
- ✓ Choose between choropleth, dot density, proportional symbol, and cartogram maps
- ✓ Apply normalization to avoid the choropleth trap in geographic visualization
- ✓ Apply cognitive load principles (Miller's Law) and eye-tracking patterns to dashboard design

Scatterplots & Relationships

Understanding correlations through coordinate geometry | Slides 3–12

The Scatterplot

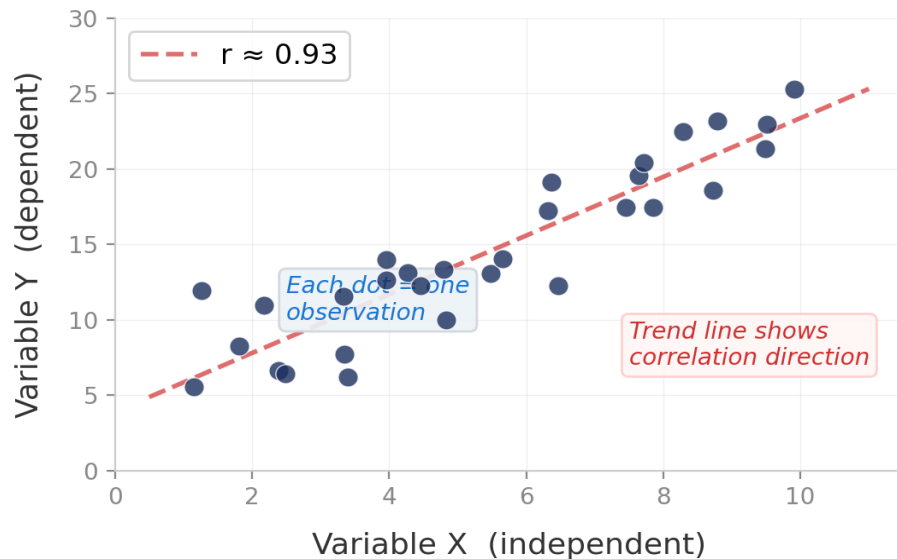
A scatterplot places individual data points on a two-dimensional plane where the horizontal axis represents one variable and the vertical axis represents another.

Each point's position reveals the relationship between those two measurements.

Definition

Using Cartesian coordinates (x, y) to display values for two variables simultaneously. Position encodes value.

Scatterplot: Position Encodes Value



Discussion: Draw a scatterplot on the whiteboard with student-contributed height vs. shoe size data.

Why Use Scatterplots?

Finding "the story" between variables

Reveal Relationships

See if two variables move together, apart, or independently

Spot Outliers

Identify data points that deviate from the overall pattern

Guide Hypotheses

Generate research questions: Does X really cause Y?

Assess Model Fit

Visually check if a linear or nonlinear model is appropriate

Exercise: Brainstorm 3 variable pairs you think are correlated.

Anatomy of a Scatterplot

1

X-Axis (Horizontal)

The independent or explanatory variable

2

Y-Axis (Vertical)

The dependent or response variable

3

Data Points (Glyphs)

Each dot = one observation; shape/color can encode a third variable

4

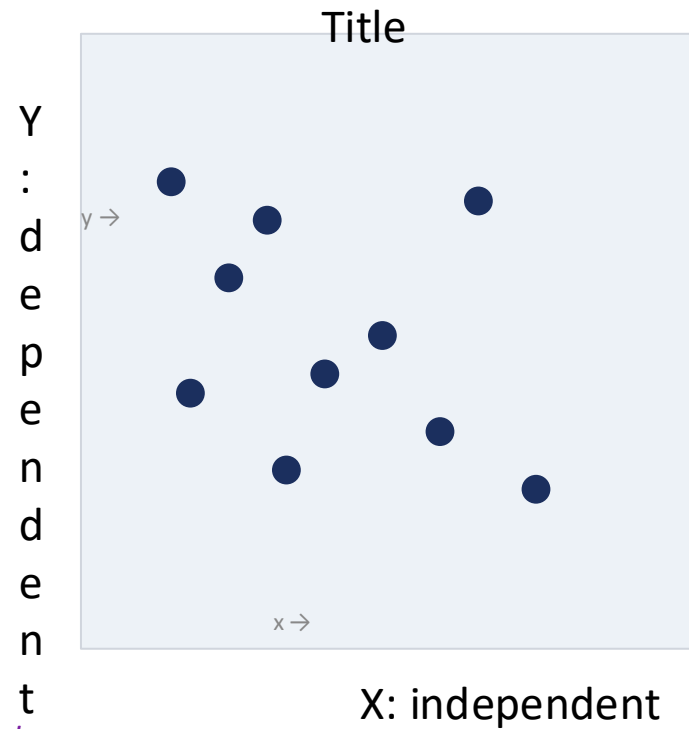
Scale & Gridlines

Tick marks for reference; gridlines aid reading exact values

5

Axis Labels & Title

Always include units (e.g., "Height (cm)") for clarity



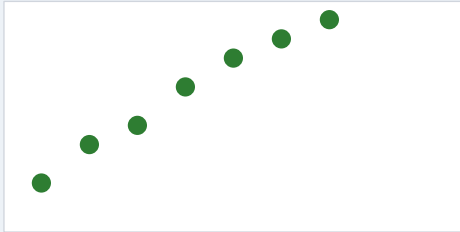
Note: A scatterplot without labels is just modern art — beautiful but meaningless.

Correlation Types

The three patterns you will see in scatterplots

Positive Correlation

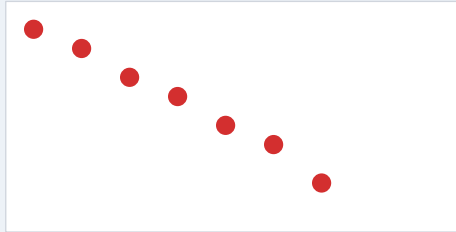
As $X \uparrow$, $Y \uparrow$



Study hours $\rightarrow$ Exam score

Negative Correlation

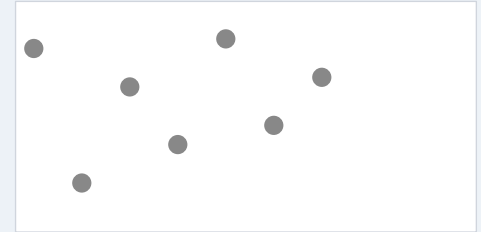
As $X \uparrow$, $Y \downarrow$



Car age $\rightarrow$ Resale value

No Correlation

No systematic pattern



Shoe size $\rightarrow$ IQ

The Math of Scatterplots

Pearson Correlation Coefficient (r)

Pearson Correlation Coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2]}}$$

$$r = +1.0$$

Perfect positive

$$r = 0$$

No linear correlation

$$r = -1.0$$

Perfect negative

⚠ Caution: r only measures linear relationships.

A dataset with $r \approx 0$ may still have a strong nonlinear pattern (quadratic, sinusoidal). Always plot first!

Example: Study Hours vs. Exam Scores

Student	Hours (X)	Score (Y)
A	2	58
B	4	65
C	5	72
D	6	78
E	8	85
F	10	92

Result: $r \approx 0.99 \rightarrow$ Very strong positive correlation. More study hours strongly predict higher scores.

Exercise: Compute r by hand for the first 4 students, then check with a calculator for all 6.

Overplotting: The 'Blob' Problem

What happens when you have 10,000+ data points?

✗ The Problem

- Points overlap into a solid mass
- Individual observations disappear
- Density patterns become invisible
- Outliers are buried

✓ Consequences

- Hidden density: Can't see where data concentrates
- Lost outliers: Unusual points invisible
- False uniformity: Different-density areas look the same
- Wasted ink: Every pixel is black — no information gradient

Discussion: What solutions can you think of?

Solution 1: Alpha Blending

Set each point's opacity to a low value (e.g., $\alpha = 0.05$). Where many points overlap, the color builds up — revealing density without hiding structure.

Rule of Thumb

$\alpha \approx 1 / \text{max_overlapping_points}$

100 pts $\rightarrow$ 0.3–0.5 | 1K $\rightarrow$ 0.1 | 10K $\rightarrow$ 0.02 | 100K+ $\rightarrow$ 0.005

Python Code

```
plt.scatter(x, y,  
            alpha=0.05,  
            s=5,  
            c='teal')
```

Live demo: Toggle alpha from 1.0 down to 0.01 in a Jupyter notebook. Watch density emerge.

Solution 2: Jittering

When one or both variables are discrete (e.g., ratings 1–5, grade levels), many points land on identical coordinates. Jittering adds tiny random offsets so you can see each observation.

Python Code

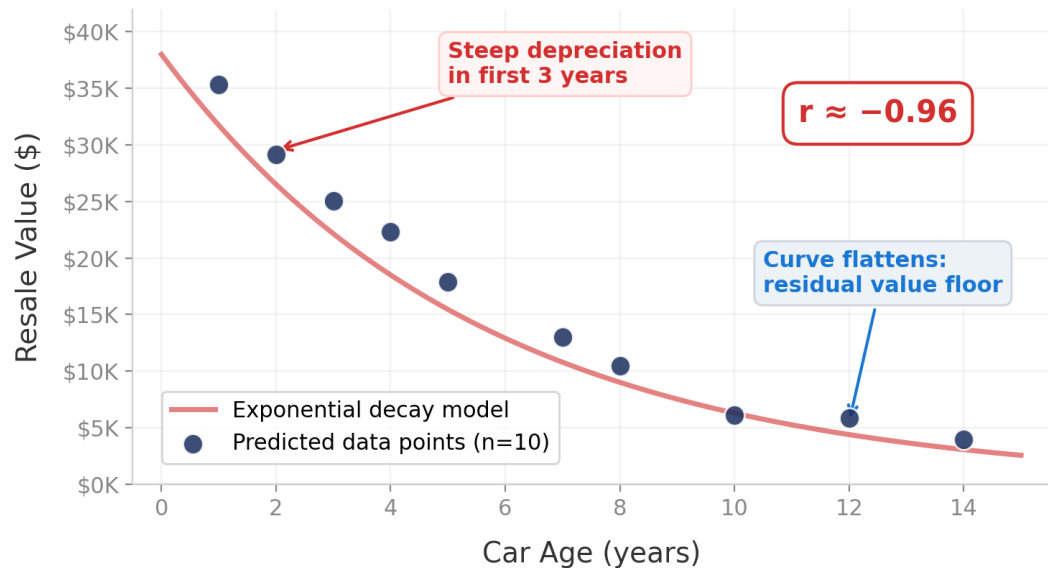
```
plt.scatter(  
    x + np.random.normal(0, 0.1,  
    len(x)),  
    y, alpha=0.5)
```

Pen & Paper Exercise: Car Age vs. Resale Value

1. Draw X-axis: Car Age (0–15 years). Y-axis: Resale Value (\$0–\$40,000).
2. Predict the correlation direction. (Negative? Linear or curved?)
3. Would jittering be needed? (No — both variables are continuous.)
4. Estimate Pearson r . Draw 10 predicted data points on your graph.
5. Compare with a partner. Then we'll show real Kelley Blue Book data.

Exercise Answer: Car Age vs. Resale Value

Car Age vs. Resale Value — Predicted Scatterplot



Answers

1. Axes

X: Car Age (0–15 yrs)

Y: Resale Value (\$0–\$40K)

2. Correlation

Negative & curved
(exponential decay, not linear)

3. Jittering?

No — both variables are continuous, no overplotting

4. Pearson r

$r \approx -0.96$ (strong negative)

Note: r underestimates the true nonlinear relationship

5. Key Insight: Real Kelley Blue Book data confirms the exponential decay pattern. A new \$38K car loses ~30% of its value in the first 2 years, then depreciation slows. Pearson r captures direction and strength, but since the relationship is curved (not linear), r^2 understates the true predictability. A log-transform or exponential model fits better.

Heatmaps & Density

Color-encoding data values for pattern discovery | Slides 13–24

What is a Heatmap?

A graphical representation where values are depicted by color

Core Concept

A heatmap uses a matrix layout where each cell is colored according to its value. The visual system processes color patterns faster than reading numbers—making it ideal for spotting clusters, gradients, and anomalies in large datasets.

Key advantage: Pre-attentive processing of color allows near-instant pattern detection.



Heatmap Use Cases

Where heatmaps shine across industries

Website Analytics

Click heatmaps showing where users interact most on a webpage

Biology / Genomics

Gene expression levels across samples (microarray data)

Finance

Correlation matrices between stock returns

Sports Analytics

Shot charts showing scoring probability by court/field position

Calendar Patterns

GitHub contribution grids, hourly email volume by day of week

Manufacturing

Quality control metrics across production lines and time periods



Ask each student group to identify one heatmap they've encountered. Discuss what made it effective or confusing.

Color Theory for Heatmaps

Sequential vs. Diverging color scales

Sequential

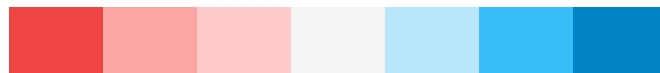
Light → Dark for data ranging from low to high (one direction). Best for counts, temperatures, or frequencies.



Low → High

Diverging

Two hues diverge from a neutral midpoint. Best for data with a meaningful center (e.g., temperature anomaly, profit/loss).



Negative ← Neutral → Positive



Rule of thumb: Use sequential for "how much" and diverging for "compared to what" questions.



Print color swatches. Have students sort them into sequential vs. diverging groups, then discuss edge cases.

The Data Matrix

How rows and columns form the heatmap grid

Matrix Structure

A heatmap is fundamentally a colored table:

- Rows = one categorical variable (e.g., genes, products)
- Columns = another variable (e.g., samples, time)
- Cell value = the measurement to be color-encoded

The matrix can be reordered (sorted or clustered) to reveal hidden patterns.

	Mon	Tue	Wed	Thu	Fri
9AM	2	5	3	7	4
12PM	6	8	7	9	5
3PM	4	6	5	8	3
6PM	3	4	2	5	2



Provide a spreadsheet of numbers. Ask students to manually color cells using 3 colors before showing the software version.

Clustering in Heatmaps

Using dendrograms to group similar rows and columns

Why Cluster?

Raw heatmaps can look noisy. Hierarchical clustering reorders rows and columns so that similar profiles appear adjacent. This reveals blocks of shared behavior.

Dendrogram

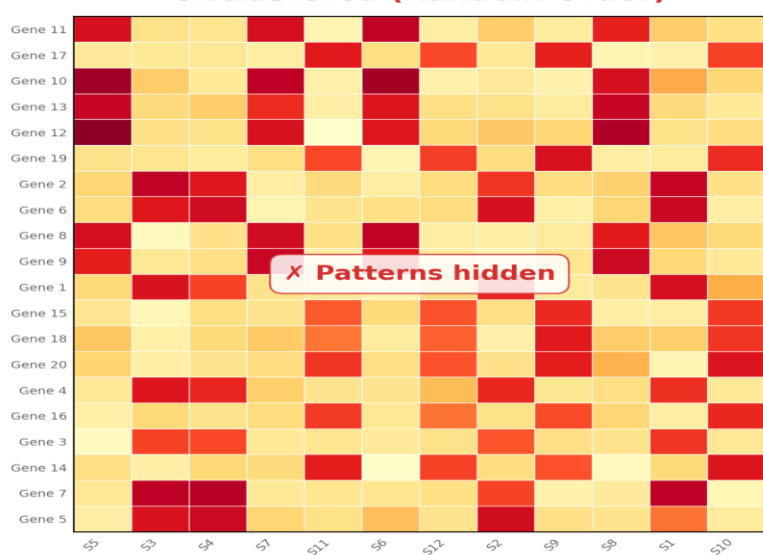
A tree diagram attached to the heatmap showing the merge order. Items that join early are most similar. Cut the tree at any height to form K clusters.

Python: `seaborn.clustermap()`

```
import seaborn as sns
sns.clustermap(
    data,
    method='ward',
    cmap='viridis',
    figsize=(8, 8)
)
```

Clustered vs. Unclustered Heatmap

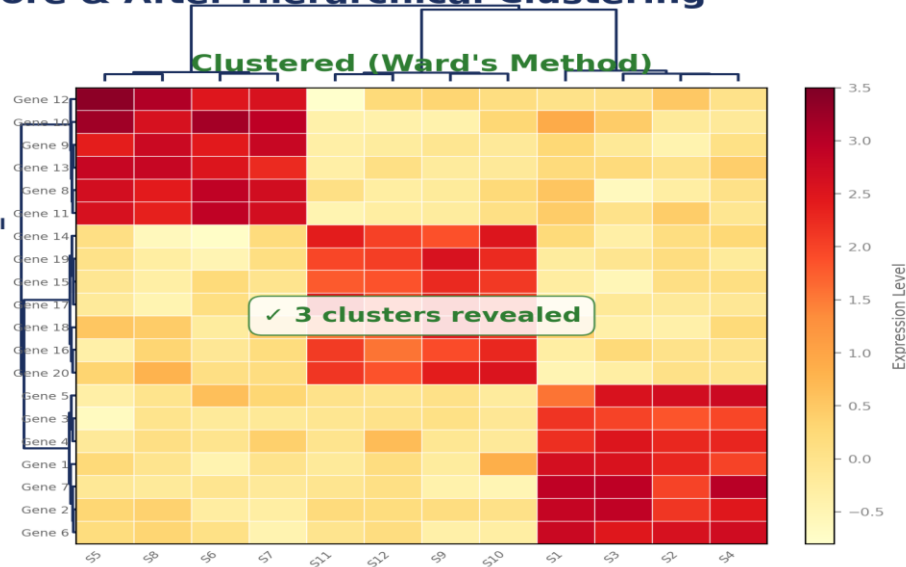
Gene Expression Heatmap: Before & After Hierarchical Clustering



Hierarchical Clustering



(Ward's method)



✗ Random row/column order

Patterns are invisible. The same data looks like noise when genes and samples aren't ordered by similarity.

✓ Hierarchical clustering

Ward's method groups similar rows & columns together, revealing 3 distinct gene expression clusters as diagonal blocks.

Dendrograms show distance

Tree-shaped branching diagrams on top and left edges encode how similar each gene/sample pair is — shorter branches = more similar.

Spatial Heatmaps

Mapping density over a 2D floor plan or map

Unlike matrix heatmaps, spatial heatmaps overlay color onto a physical layout (a floor plan, a field, or a map). The x/y positions correspond to real-world locations, and color intensity represents density or magnitude at each point.

Retail Store

Foot traffic density across aisles and departments

Sports Field

Shot attempts or pass completions by field position

City Map

Crime incidents, taxi pickups, or WiFi signal strength

Example: The GitHub Contribution Grid

Analyzing time-based patterns in a calendar heatmap

A calendar heatmap maps activity over time. GitHub's contribution grid uses columns for weeks, rows for days, and green intensity for commit count.



Patterns to look for:

- Weekend vs. weekday activity patterns (work-life balance indicator)
- Streaks and gaps (motivation cycles, project deadlines)
- Seasonal trends (semester schedules, holiday effects)



Sketch your own 'study contribution grid' for the past month. Discuss what patterns emerge.

2D Binning: Hexbins & Rectangular Bins

Converting overplotted scatterplots into density displays

How 2D Binning Works

1. Divide the 2D space into bins (hexagons or rectangles)
2. Count the points falling in each bin
3. Color each bin by its count

This combines the position information of scatterplots with the color encoding of heatmaps.

Hex vs. Rectangle

Hexagons are preferred because:

- Each hex has 6 equidistant neighbors (vs. 4+4 for squares)
- They tile without diagonal bias
- They produce smoother visual gradients

Python: `plt.hexbin(x, y, gridsize=30)`

Kernel Density Estimation (KDE)

Smoothing discrete points into a continuous density surface

KDE Formula (Gaussian Kernel)

$$\hat{f}(x) = (1/nh) \sum K((x - x_i)/h) \quad \text{where} \quad K(u) = (1/\sqrt{2\pi}) \exp(-u^2/2)$$

Key Parameter: Bandwidth (h)

- Small $h \rightarrow$ spiky, noisy (overfitting)
- Large $h \rightarrow$ smooth, blurry (underfitting)
- Optimal h balances bias and variance

2D KDE $\rightarrow$ Contour Plot

Extending KDE to 2 dimensions produces a smooth surface. Displayed as a contour plot or filled contour, it reveals the underlying distribution shape. Python:

```
sns.kdeplot(x=x, y=y, fill=True)
```

The "Rainbow Map" Pitfall

Why the Jet/Rainbow color scale is misleading



Problems with Rainbow

- Not perceptually uniform: equal data steps $\neq$ equal perceived color steps
- Creates false boundaries: sharp yellow-green and cyan-blue transitions create phantom contours
- Inaccessible: ~8% of men are red-green colorblind
- Luminance inversions: bright yellow appears 'higher' than dark blue regardless of actual value
- Hides patterns: smooth gradients become banded noise



Better Alternatives

Viridis, Magma, Plasma, Inferno: These palettes increase in luminance monotonically, work in grayscale, and are colorblind-safe.



Viridis



Magma

Perceptually Uniform Palettes

Viridis, Magma, and designing for accessibility

What makes a palette 'perceptually uniform'?

A fixed step in data value always produces the same perceived change in color. This means the visualization faithfully represents the data—no artificial banding, no phantom features. These palettes are designed to work in grayscale printing and for all forms of color vision deficiency.



Pen & Paper Exercise

Given this 4×4 grid of values:

[2, 5, 8, 3] [7, 9, 4, 6] [1, 3, 7, 8] [5, 2, 6, 9]

Color-code them into a 3-step sequential heatmap (light, medium, dark). Values 1–3 = light, 4–6 = medium, 7–9 = dark.

MODULE 3

Spatial Layouts & Non-Geographic Maps

Using position to encode meaning beyond coordinates | Slides 25–36

Spatial Encoding

The x-y plane doesn't have to represent numbers. Position can encode:

- Category membership (treemaps)
- Connections (network graphs)
- Comparison (small multiples)
- Cyclic time (polar coordinates)

The choice of spatial layout is a DESIGN DECISION, not a mathematical one. Different layouts emphasize different aspects of the same data.

Discussion: Match datasets to layout types. A) Budget breakdown B) Social connections C) Sales by region D) Trade flows

Spatial Encoding

Moving beyond x and y — using position to imply meaning

In 2D visualization, position is the most powerful visual channel. But x and y don't have to represent numeric variables — they can encode categorical groupings, hierarchies, connections, or spatial metaphors. This module explores layouts where position itself tells the story.



Rectangular

Treemaps, partition maps



Circular

Chord diagrams, polar coords



Force-Directed

Network graphs, clusters



Grid/Small Multiples

Trellis plots, facets

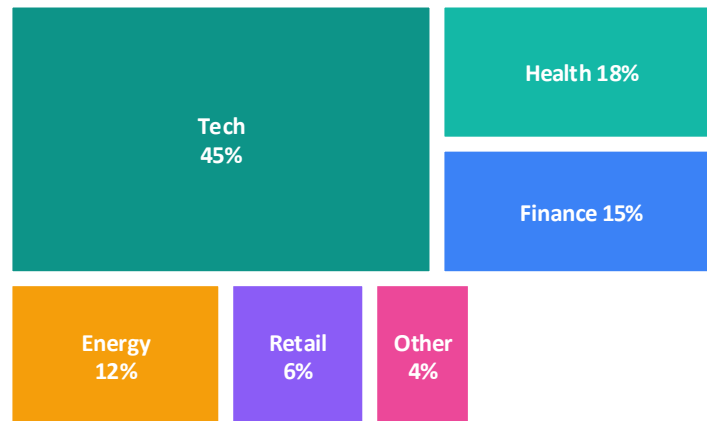
Rectangular Layouts: Treemaps

Area-based visualization of hierarchical and part-to-whole data

How Treemaps Work

Each rectangle's area is proportional to a quantitative variable (e.g., revenue, file size). Nesting shows hierarchy: a folder contains files, a category contains subcategories.

Invented by Ben Shneiderman in 1990 at the University of Maryland.



Circular Layouts

Chord diagrams and polar coordinates

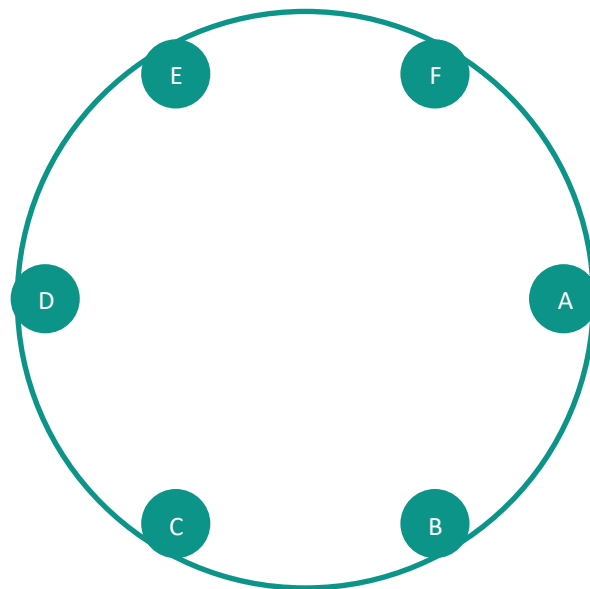
Chord Diagrams

Show flows or connections between entities arranged in a circle. Arc width encodes the strength of the connection. Best for showing who-connects-to-whom patterns.

Examples: Trade between countries, migration flows, interdepartmental collaboration.

Polar Coordinates: (r, θ)

Used in radar/spider charts, wind roses, and clock-based displays. Natural for cyclic data (hours, months, compass directions).



Small Multiples (Trellis Plots)

Repeating the same spatial layout across different categories

Same structure, different data subsets. The viewer's eye compares across panels, spotting differences instantly. Edward Tufte calls them 'the best design solution for a wide range of problems.'

Design Rules

- Keep axes identical across all panels
- Use consistent color encoding
- Order panels meaningfully (alphabetical, temporal, or by effect size)

Group A



Group B



Group C



Group D



Force-Directed Layouts

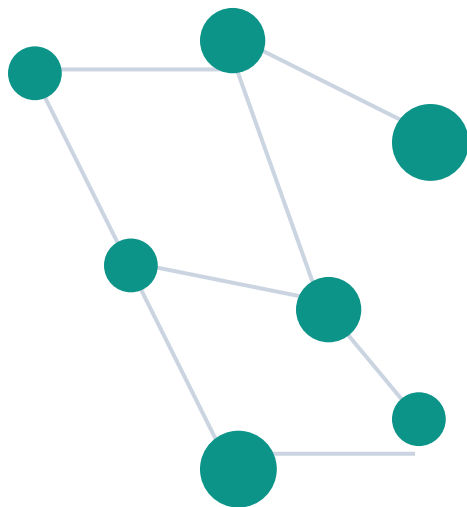
Position based on attraction and repulsion (network spatiality)

The Physics Metaphor

Nodes are treated as charged particles (repulsion) and edges as springs (attraction). The algorithm iterates until the system reaches equilibrium — nodes that share many connections cluster together naturally.

Key parameters:

- Repulsion strength: How far nodes push each other apart
- Spring constant: How tightly connected nodes pull together
- Iterations: More iterations → more stable layout



Aspect Ratio Matters

How width-to-height ratio changes our perception of slope and trend

The same data plotted at different aspect ratios can make a trend look dramatic or flat. Cleveland's 'banking to 45°' principle suggests choosing an aspect ratio where the average slope of line segments is about 45 degrees for optimal perception.

Wide (4:1)

Trend looks flat, changes seem minor



Tall (1:2)

Same data looks like a dramatic spike



Coordinate Systems

Cartesian (x, y) vs. Polar (r, θ)

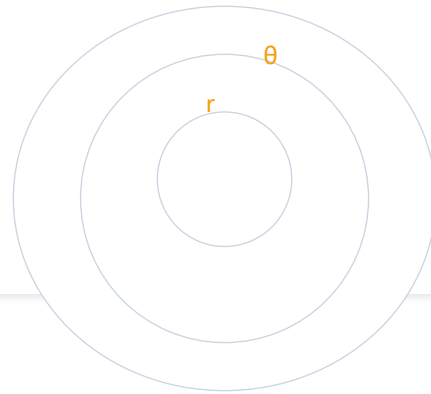
Cartesian

Position defined by perpendicular axes (x, y).
Intuitive for most data. Good for: scatterplots, line charts, bar charts, heatmaps.



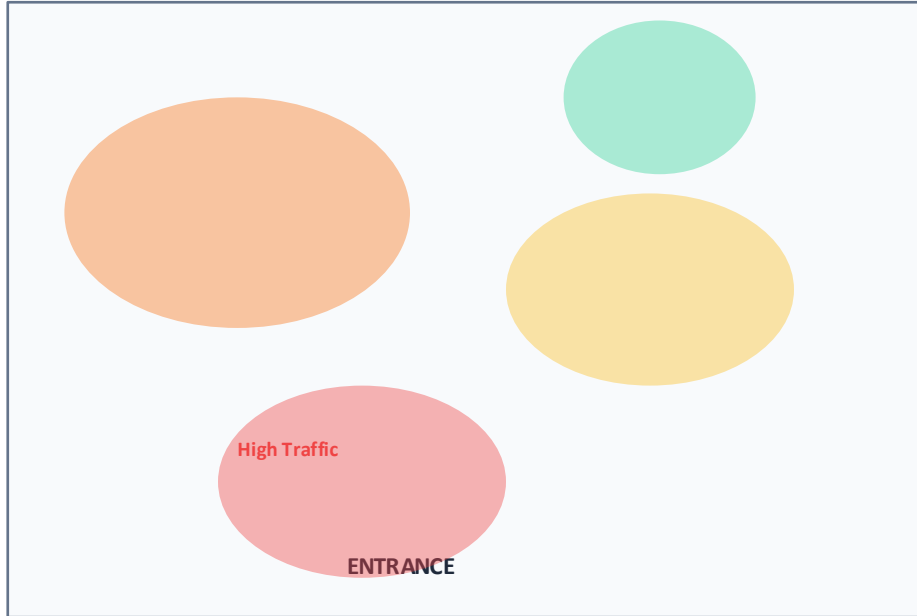
Polar

Position defined by distance (r) and angle (θ) from a center point. Natural for cyclic or directional data.
Good for: wind roses, radar charts, clock visualizations.



Example: Retail Store Foot Traffic

A floor plan heatmap showing customer movement patterns



Analysis

- Entrance zone has highest density (expected)
- Left wing draws more traffic than right
- Back-right corner is a 'dead zone'
- Use this data to optimize product placement and signage

Dashboard Layouts & Eye Tracking

The Z-pattern and F-pattern of information scanning

Z-Pattern

For dashboards and pages without heavy text.
The eye moves: top-left → top-right → diagonal to bottom-left → bottom-right.

Place the most important KPI in the top-left. Put the call-to-action in the bottom-right.



Dashboard Grid Rules

- Top row: KPIs and summary metrics (big numbers)
- Middle: Main visualization (the largest chart)
- Bottom: Detail tables or secondary charts
- Left sidebar: Filters and navigation
- Limit to 5–7 visual elements per screen

Cognitive Load & Miller's Law

How many spatial elements can a human process simultaneously?

7 ± 2

Miller's Magic Number

The average person can hold 5–9 distinct items in working memory. Exceed this and comprehension drops sharply.

Design Implications

- Limit legend categories to 5–7 colors
- Group related elements visually
- Use progressive disclosure for complex views
- Chunk information hierarchically

Grids & Alignment: The Rule of Thirds

Divide your visualization canvas into a 3×3 grid. Place key elements at intersection points for visual balance. This principle from photography applies directly to dashboard and infographic design.

Exercise: Smart Home Control Panel

Design a 2D layout showing:

- a) Room temperature for 5 rooms (spatial heatmap)
- b) Energy consumption over 24 hours (line/area chart)
- c) Device status for 8 devices (on/off indicators)
- d) Security camera feeds (4 thumbnails)

Apply: Z-pattern, Miller's Law (≤ 7 groups), Rule of Thirds.

Time: 15 min in pairs. Each pair presents (1 min). Class votes on best design.

Homework: Compare your design to a real smart home dashboard (Home Assistant). What would you change?

MODULE 4

Geographic Visualization

From latitude/longitude to meaningful spatial stories | Slides 37–50

Introduction to GIS

What makes Geographic Information Systems different?

GIS combines spatial data (where things are) with attribute data (what things are). Unlike regular charts where x and y are abstract variables, in GIS they represent real locations on Earth's surface. This adds unique challenges: projections, coordinate systems, and spatial relationships.

Spatial Queries

Find all hospitals within 5km of a flood zone

Overlay Analysis

Combine soil type, rainfall, and elevation layers

Geocoding

Convert addresses to lat/lon coordinates



Discuss: what can GIS do that Google Maps cannot? (Spatial analysis)

Latitude and Longitude

The spherical coordinate system for Earth

Latitude (φ)

Angle north or south of the Equator.

Range: -90° (South Pole) to $+90^\circ$ (North Pole)

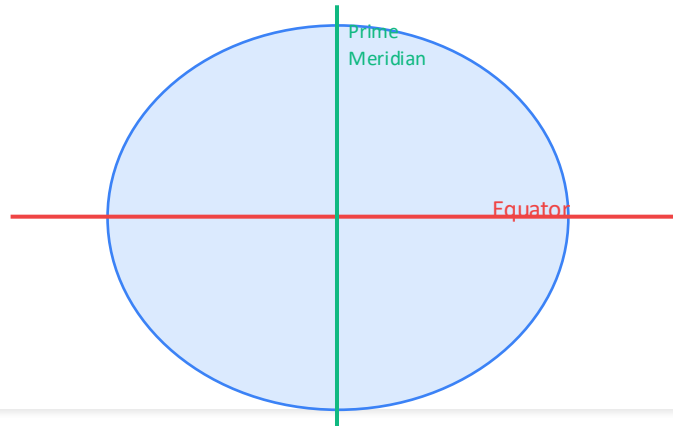
Lines of latitude run east-west (parallels).

Longitude (λ)

Angle east or west of the Prime Meridian.

Range: -180° (west) to $+180^\circ$ (east)

Lines of longitude run north-south (meridians).



Euclidean Distance (2D)

$$d(p, q) = \sqrt{ (q_1 - p_1)^2 + (q_2 - p_2)^2 }$$

Map Projections: The "Orange Peel" Problem

You can't flatten a sphere without distortion

Imagine peeling an orange and pressing it flat. The peel tears and stretches. Every map projection makes a trade-off: it can preserve area, shape, distance, or direction — but never all four simultaneously.

Mercator

Preserves: Shape & angles

Distorts: Area (Greenland = Africa?!)

Use case: Navigation, web maps

Peters

Preserves: Area (equal-area)

Distorts: Shape (stretched/squished)

Use case: Social justice maps

Robinson

Preserves: Compromise (neither)

Distorts: Mild everywhere

Use case: General reference atlases

Choropleth Maps

Shading regions (states, countries) based on data values

How Choropleths Work

Each geographic region is colored according to a data variable (e.g., unemployment rate, median income). The map reader compares shading intensity across regions to identify spatial patterns.

Most common map type in news media and government reports.



Light = Low Dark = High



What patterns jump out? What might be misleading?

The Choropleth Trap

"Land doesn't vote, people do" — Population density bias



The biggest danger of choropleth maps: large, sparsely populated regions dominate the visual even though they represent fewer people. Small, dense urban areas are almost invisible despite containing millions.

Problem Example

A US election map colored by county: massive red/blue areas in rural regions suggest dominance, but tiny colored dots in cities may represent far more voters.

Solutions

- Use cartograms (resize regions by population)
- Switch to dot density maps
- Normalize by population (per capita)
- Use proportional symbol maps instead

Normalizing Data for Maps

Why "Per Capita" beats "Total Count"

Raw Counts Are Misleading

Mapping raw totals essentially maps population. California will 'lead' in almost every total count metric simply because it has 39 million people.

Total crimes, total COVID cases, total GDP — all dominated by the most populous regions.

Normalized Rates Reveal Truth

Dividing by population produces rates that are comparable across regions of different sizes.

$$\text{Rate} = (\text{Count} / \text{Population}) \times 100,000$$

Now a small state with high incidence stands out, and a large state with normal incidence fades to neutral.

Normalization Formula

$$\text{Rate per 100k} = (\text{Observed Count} / \text{Population}) \times 100,000$$

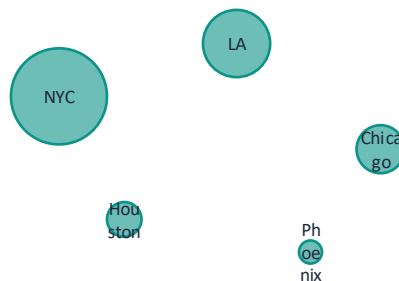
Proportional Symbol Maps

Using sized symbols (bubbles) overlaid on a geographic base map

How They Work

Place a symbol (usually a circle) at each location. The symbol's area is proportional to the data value. This avoids the choropleth trap because visual weight corresponds to data magnitude, not geographic area.

Important: Scale by area (value), not radius, to avoid exaggerating differences.



Dot Density Maps

One dot = X amount of people, events, or things

Design Principles

- Each dot represents a fixed quantity (e.g., 1 dot = 500 people)
- Dots are placed randomly within regions, giving a sense of distribution
- No choropleth trap: dots naturally cluster where density is high
- Can use multiple colors for different categories (e.g., racial dot maps)
- Key decision: choosing the dot-value ratio and dot size



Cartograms

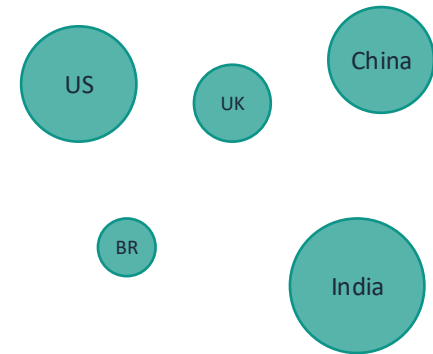
Distorting geography to represent data size

Reshaping the Map

A cartogram resizes each geographic region so its area is proportional to a data variable (e.g., population, GDP). Countries or states 'inflate' or 'deflate' based on their data values.

Types: Contiguous (stretchy), Non-contiguous (separated), Dorling (circles replace regions).

Dorling Cartogram



Trade-off: Cartograms sacrifice familiar geography for honest data representation. Readers may not recognize distorted shapes.

Layers in Spatial Data

Base map, raster layer, vector layer

Layers stack to create the final map ↓

Base Map

The geographic reference: streets, terrain, or satellite imagery. Provides spatial context without carrying data.

Raster Layer (Pixels)

Grid of cells where each pixel has a value: temperature, elevation, land cover classification. Think satellite imagery or weather radar.

Vector Layer (Points/Lines/Polygons)

Discrete features: city points, road lines, country polygons. Each feature has geometry + attribute data (name, population, etc.).

Tools for Geographic Visualization

From no-code to full programming flexibility

Tableau / Power BI

No-Code

- + Drag-and-drop, beautiful defaults
- Limited projection control

Python: Folium

Low Code

- + Interactive web maps, Leaflet-based
- Requires basic Python

Python: GeoPandas

Medium Code

- + Full spatial analysis + matplotlib
- Steeper learning curve

R: sf + ggplot2

Medium Code

- + Elegant grammar of graphics for maps
- R-specific ecosystem

QGIS

Desktop GIS

- + Full GIS capabilities, free & open
- Complex interface

D3.js / Mapbox

Advanced

- + Total creative control, web-native
- Significant coding required

Choosing the Right 2D Spatial Encoding

A decision framework for your data type

Data Type	Best Encoding	Avoid
Two numeric variables	Scatterplot / Hexbin	Pie chart
Matrix of values	Heatmap	3D surface (hard to read)
Hierarchical (part-to-whole)	Treemap / Sunburst	Stacked area
Network / connections	Force-directed graph	Table of edges
Data by geographic region	Choropleth (normalized!)	Unnormalized totals
Point locations on Earth	Dot density / Proportional symbol	Choropleth of counts
Cyclic / directional data	Polar / Wind rose	Cartesian line chart

Final Exercise: Choropleth Comparison

MAP A: Total COVID Cases by State

- Raw count visualization
- California, Texas, Florida dominate
- Essentially maps population
- Dramatic but misleading

MAP B: Cases per 100,000 Population

- Normalized rate visualization
- Reveals true severity across states
- Small states may rank highest
- Honest and actionable

Answer on paper:

1. Which map is more useful for public health decisions? Why?
2. Identify a state that ranks high on Map A but low on Map B. Explain.
3. What normalization formula converts Map A into Map B?